Capture SOP – scanning a room for Namma Space

Team Triple Musketeer · Techfest IIT Bombay 2026–27 · Standard operating procedure, version of 27 October 2026

Who this is for: anyone with a smartphone. No measuring tape, markers or 3D skills are needed. **Result:** a walkable 3D model of the room, viewable in any web browser.

Route	Phone	Time in the room	You get
A. Polycam scan (recommended)	iPhone / iPad with LiDAR	5–10 min	Model in real metres + photoreal view
B. Photos or video	Any phone	10–15 min	Model (set its size with one known length)

> Route A was used for the team's own room (iPhone 14, 25 October 2026, 218 photos) and the lessons from that scan are written into it. Route B was developed and measured on public datasets (ETH3D, ARKitScenes, TUM).

1. Before you start – both routes (2 minutes)

Do	Why
Switch on all lights; open curtains only if the windows are not in view for long	Dark images are noisy and blurry; bright windows blind the camera
Clean the lens	A smudge blurs everything
Move people and pets out of the room; do not move furniture during the scan	Moving things cannot be reconstructed
Hide personal documents, screens and family photos	The model is photographic and may be published
Charge the phone and free some storage (≥ 2 GB)	A full scan with photos is a few hundred MB

2. Route A – Polycam scan (recommended)

Settings: Polycam → **LiDAR** mode (Room / Space capture).

How to move. The photoreal view is learned from the photos Polycam takes while you scan, so *what the phone looks at* decides the quality.

1. Start in a corner, phone at chest height, and walk **slowly** (about 30 cm per second) along the walls.
2. **Point the phone straight at each wall from about 1–1.5 m** and walk along it – including plain white walls and the wall beside every door. *Lesson from our room: a wall the phone only saw from across the room or at a steep angle came out smeared up close, even though its shape was right.*
3. Do one lap with the phone **tilted down**, so the **floor** near the walls and around furniture is covered (needed for walking and routes), and one lap **tilted up** for the line where walls meet the ceiling.
4. Go **around** large furniture (bed, wardrobe, table), not just past it, so it is seen from several sides.
5. **Never turn on the spot** and never whip the phone around; keep each part of the room in view for a few seconds. Do not point into a bright window for long.
6. Finish, let Polycam process, and check its preview for holes. Rescan a missed area **before leaving the room**.

Export (both are needed):

- Export → **GLTF** – the 3D model.
- Export → **Images** – the keyframe photos with camera positions, needed for the photoreal view.

Save both to Files or AirDrop them to the computer.

Into Namma Space: in the Studio choose **New space from Polycam** and drop both files (or scan the Studio's QR code with the phone → **Import a Polycam export**). The import repairs the scan automatically (wavy ceiling, holes, hanging debris) in a few seconds. Then, on the space's card, click **Make the photoreal view** (needs an NVIDIA GPU; about 20–25 minutes for ~200 photos). The model is in real metres from the phone's tracking: check one known length (a door width) with the viewer's measuring tool (**M**).

3. Route B – photos (any phone)

Phone settings: main 1× camera, no zoom, same phone for the whole room; hold the phone **sideways (landscape)** for every photo; iPhone: Settings › Camera › Formats › **Most Compatible** (JPEG – HEIC photos are refused).

1. **Start at a corner.** Stand about one step in from the wall and face *into* the room.

2. **Walk along the walls.** Take a photo, take **one small step (about 0.5 m)** sideways, take the next photo. Consecutive photos should share about **two thirds** of what they show.
3. **Tilt slightly down** so the **floor is in the lower third** of every photo – floors are needed for walking and were the part most often missing in our measurements.
4. **Hold still** for each shot (sharp beats many). In low light, lean on something.
5. **Go all the way around**, then do a **second, lower lap (waist height)** around large furniture so its legs and the floor around it are seen.
6. **Doorways:** take 3–4 photos while walking *through* each doorway so the rooms connect.
7. **Plain surfaces** (white walls, glass, mirrors, shiny tiles) cannot be reconstructed on their own: always include an edge, a picture, furniture or a corner in the frame.
8. **How many:** small room 40–80 photos, living room 80–150 (the Studio accepts 10–300).

Video instead of photos (only if you can keep it slow): one continuous take of **1–3 minutes**, walking **slowly (about 30 cm per second)** with the same rules; iPhone: turn **HDR Video off**. Never turn on the spot – walk a curve instead. The Studio extracts up to 300 sharp frames automatically. One video per space; do not mix photos and video. *Evidence: two public room videos recorded with fast sweeps and turns on the spot could not be reconstructed (only about a third of the frames could be placed); slow, steady photos of the same kind of room worked.*

Upload and check:

1. In the Studio, create a space and drag the photos or the video into the upload area.
2. Press **Check my capture**. Before any long processing it reports:
   - blocking problems: too few photos, unreadable files, HEIC, photos and video mixed;
   - warnings: blurry, too dark or too bright, few details, near-duplicates, photos that overlap nothing, groups of photos that do not connect.
3. Retake the flagged photos if the check suggests it, then press **Start reconstruction**.

Real-world size: photos alone do not tell a room's true size. The model is always correctly *shaped*; to show metres, open the walkthrough, choose **Measure / set scale**, click both ends of something you know (a door width, a table edge) and type its real length. This is optional and can be done at any time.

4. Checklist before leaving the room

✓	Check
☐	Every wall seen straight on from 1–1.5 m (route A), or photographed with the floor visible (route B)
☐	Floor covered around furniture; large furniture seen from several sides
☐	Doorways captured while walking through them
☐	Preview checked for holes; missed areas rescanned
☐	Route A: **both** GLTF and Images exported

5. Privacy

Everything Namma Space does runs on your own computer; nothing is uploaded unless you publish the walkthrough yourself. Polycam is a third-party app with its own account and privacy settings – check them before scanning.